

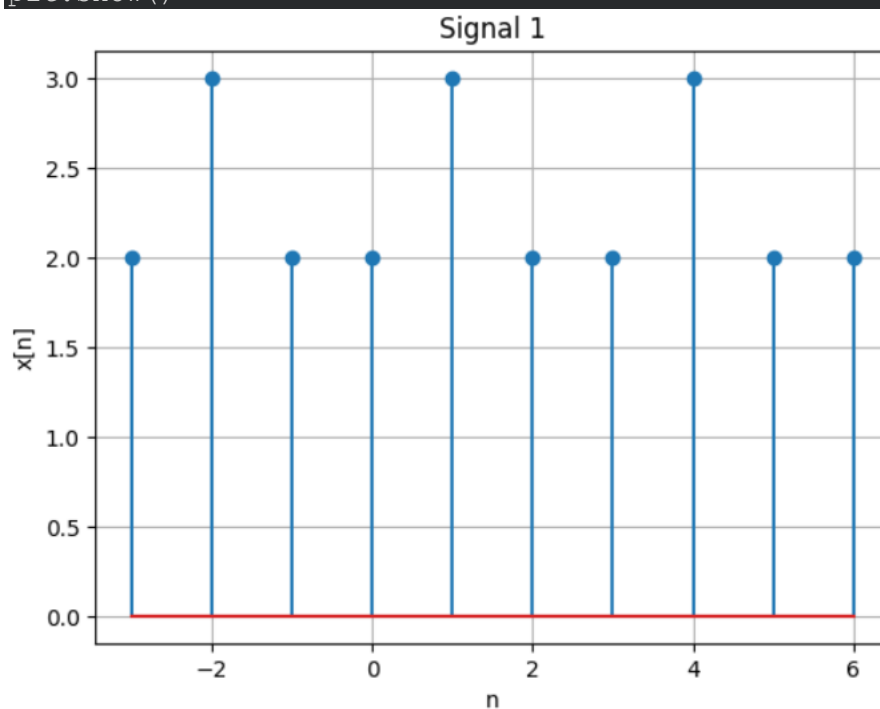
Open-ended Problem

```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-3, 7)

# Pattern: 2,3,2 repeating
pattern = [2, 3, 2]
x = [pattern[(i+3) % 3] for i in n]

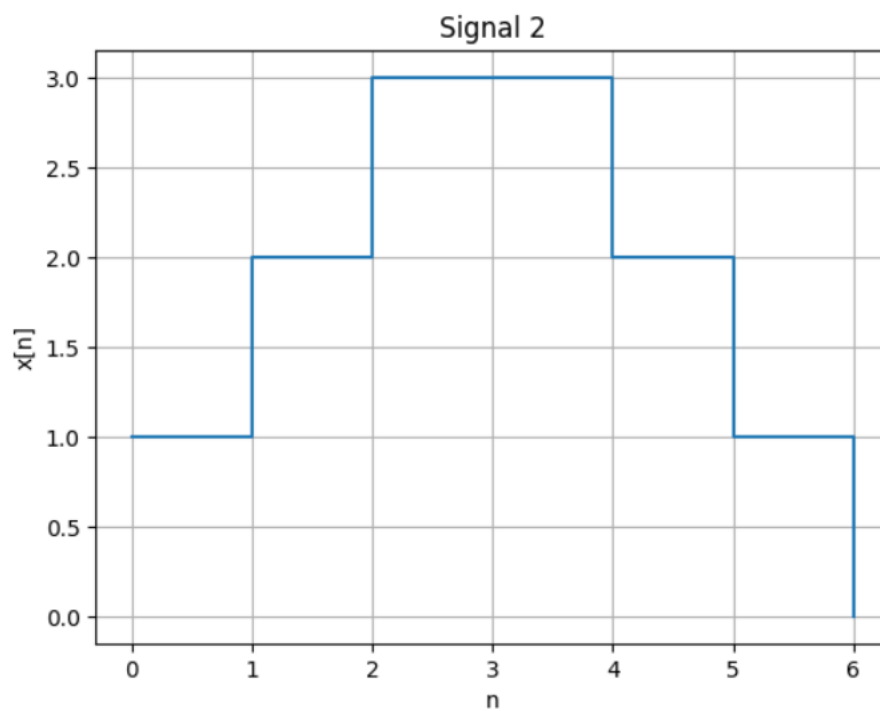
plt.stem(n, x)
plt.title("Signal 1")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```



```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(0, 7)
x = [1, 2, 3, 3, 2, 1, 0]

plt.step(n, x, where='post')
plt.title("Signal 2")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```

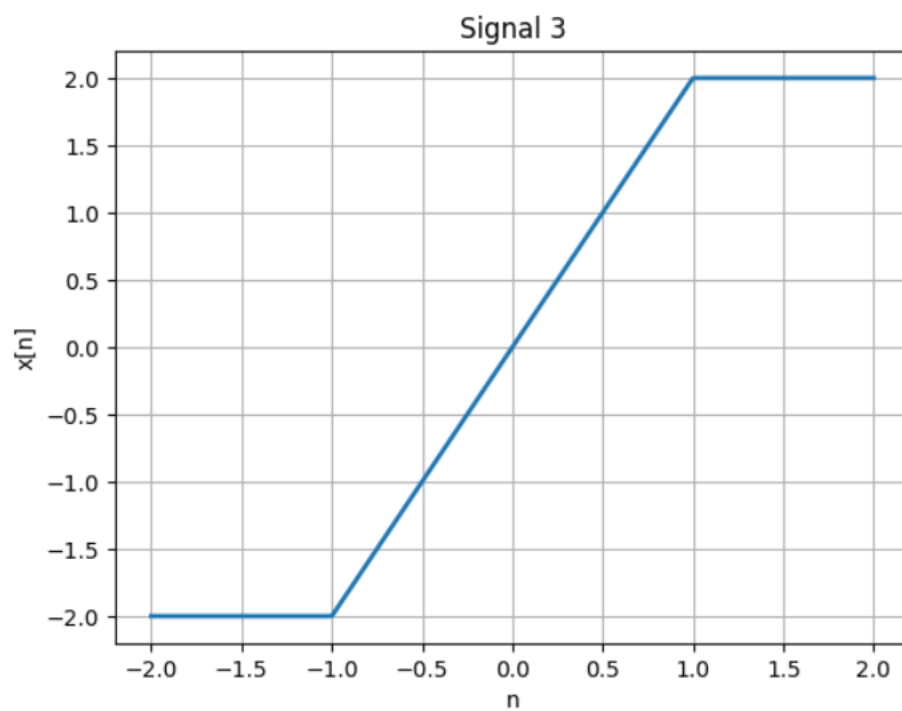


```
import numpy as np
import matplotlib.pyplot as plt

n = np.linspace(-2, 2, 400)
x = np.zeros_like(n)

x[(n >= -2) & (n < -1)] = -2
x[(n >= -1) & (n <= 1)] = 2 * n[(n >= -1) & (n <= 1)]
x[(n > 1) & (n <= 2)] = 2

plt.plot(n, x, linewidth=2)
plt.title("Signal 3")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```



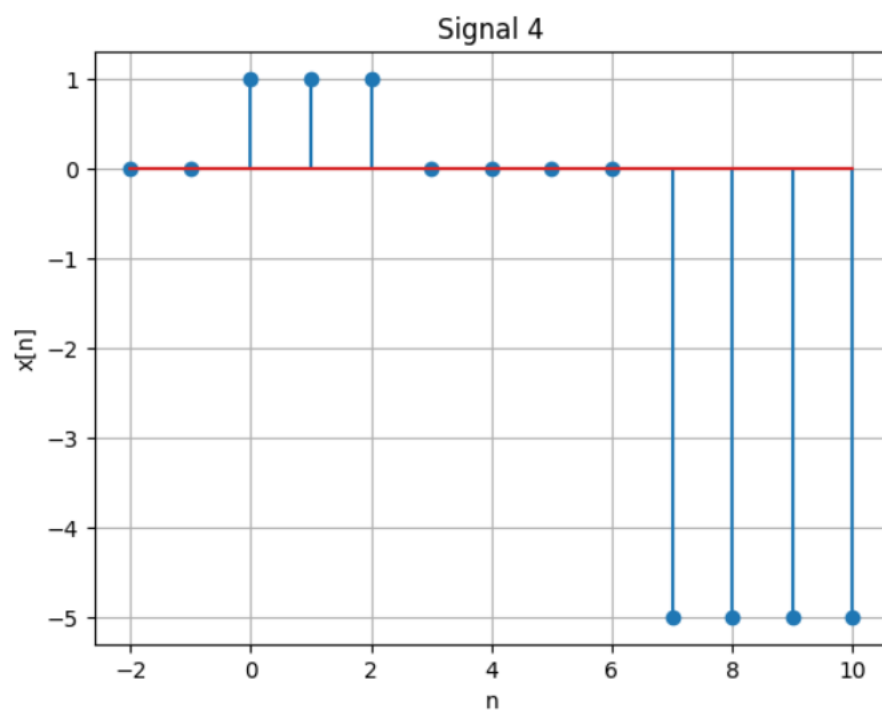
```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-2, 11)

u = lambda x: np.where(x >= 0, 1, 0)

x = u(n) - u(n-3) - 5*u(n-7)

plt.stem(n, x)
plt.title("Signal 4")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```



```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-3, 4)

delta = lambda x: np.where(x == 0, 1, 0)

x = delta(n) + 3*delta(n-1) + 5*delta(n+1)

plt.stem(n, x)
plt.title("Signal 5")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```

